SCIENCE
PASSION
TECHNOLOGY

Hops, Yeast, Pure Delight
Fermentation Temperature and Beer Aroma

Norbert Winter

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Outline

1 Motivation

2 Method

3 Results

Why Beer?

- Only four ingredients – yet thousands of aromas
- Much of the aroma is created in the fermenter
- Systematic data on fermentation temperature are missing

Research Question

How strongly does fermentation temperature shape the aroma of top-fermented beers?

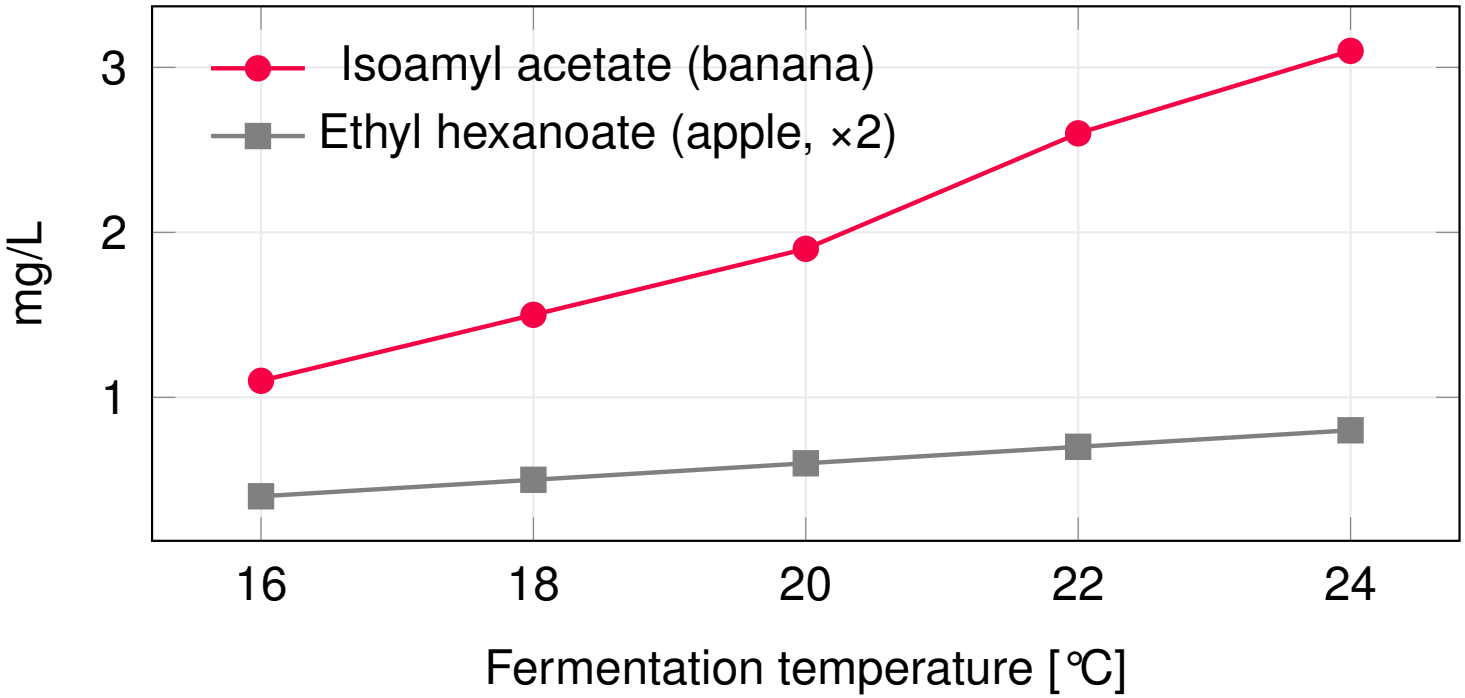
Approach

1. 12 brews of pale ale
2. 4 temperature levels (16–24 °C)
3. Gas chromatography
4. Blind tasting ($n = 24$)

Arrhenius

$$k(T) = A \cdot e^{-E_a/RT}$$

Ester Formation



Tasting

Temperature	Fruitiness	Cleanness	Overall
16 ℃	4.1	7.8	6.2
18 ℃	5.6	7.2	7.4
20 ℃	6.9	5.9	6.8
24 ℃	8.2	3.7	5.1

Conclusion

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- Next: kveik yeasts and temperature profiles

Questions?